Zihan Lin

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EDUCATION

Georgia Institute of Technology — **Atlanta, GA**
B.S. Biomedical Engineering | Robotics Minor | Expected May 2028 — Jan 2026-Present

University of Florida — **Gainesville, FL**
B.S. Biomedical Engineering | Dean's List — Aug 2024-Dec 2025

TECHNICAL SKILLS

- Programming: Python, C, MATLAB
- Robotics & Simulation: MuJoCo, URDF/MJCF, forward/inverse kinematics, trajectory generation, control systems
- **Embedded & Hardware:** STM32, CubeMX/CubeIDE, UART/TTL serial communication, stepper/servo actuation, magnetic encoders
- **Engineering & Design:** SolidWorks, Onshape, CAD modeling, rapid prototyping, sensor-actuator integration
- Research Interests: Robot manipulation, learning from demonstration, motion/path planning, surgical robotics

ROBOTICS & RESEARCH EXPERIENCE

Research on Surgical Robotics, Research Assistant — **Atlanta, GA**
BioMedical Mechatronics Lab (BM2) | Georgia Institute of Technology — Jan 2026-Present

- Developing an MRI-compatible needle-placement robotic platform, iterating mechanical architecture, patient-conforming fixture concepts, and insertion/tilt mechanisms under workspace and MRI compatibility constraints.
- Created and evaluated CAD concepts for robot structure, lead-screw/gear transmission, and rapid-prototyped components; supported design reviews focused on manufacturability and system integration.
- Built system-level understanding of surgical robotics through laboratory training and exposure to medical robotic platforms, including the da Vinci Research Kit.

6-DOF Robotic Arm — Independent End-to-End Build — **Shanghai, China**
Independent engineering project — Jun 2026-Aug 2026

- Designed and fabricated a six-axis desktop manipulator with custom 3D-printed joints, timing-belt reductions, bearing-supported shafts, modular actuator mounts, and a 24 V electrical architecture.
- Integrated STM32 firmware with stepper/servo actuation, dual absolute magnetic encoders, and UART interfaces; implemented coordinated joint commands, homing, calibration, and encoder-based closed-loop execution.
- Built a Python desktop GUI and serial command/telemetry protocol for live joint control, encoder feedback, homing, calibration, and coordinated demonstrations.
- Debugged encoder wrap-around, signed-position decoding, joint-direction, and timeout faults; added progress watchdogs and one-sided HOME safety behavior for more reliable hardware execution.
- Created a MuJoCo digital twin and Python motion pipeline using damped-least-squares inverse kinematics, multi-start search, joint-limit handling, and handwriting-style Cartesian trajectory demonstrations.

Berry-Picking Robot Simulation — **Atlanta, GA**
Georgia Tech collaborative robotics project — Jun 2026-Present

- Built and debugged MuJoCo simulation assets for a robotic berry-picking system by simplifying CAD assemblies and converting robot models across URDF/MJCF workflows.
- Diagnosed unstable joint dynamics, model constraints, coordinate-frame errors, and control-range issues to improve simulation reliability before hardware testing.

Nanomaterials Research, Research Assistant — **Gainesville, FL**
Wei Lab | University of Florida Department of Chemistry — Jan 2025-Dec 2025

- Supported research on gold nanostructures and exploratory UiO-66 photocatalysis through literature review and routine laboratory work.

Biocatalysis and Protein Engineering, Research Assistant — **Chengdu, China**
Chinese Academy of Sciences | Supervised by Prof. Wu Zhongliu — Jun 2023-Aug 2023

- Purified expressed proteins, evaluated enzymatic activity/yield using GC and LC, and verified expression through SDS-PAGE.
- Performed bacterial transformation and used AlphaFold/AutoDock for structure prediction and molecular docking.